Maxwell J. Yin

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LinkedIn | GitHub | Portfolio (Live Systems)

Summary

Ph.D. in NLP and Machine Learning Engineer specializing in LLM-powered product systems and evaluation. Experienced in designing and optimizing retrieval-augmented generation (RAG) systems, running large-scale experiments, and improving model quality, latency, and reliability in real-world applications.

Experience

Huawei Technologies - Noah's Ark Lab (Canada) 08/2025 – 02/2026

Machine Learning Engineer *Toronto, ON*

- **LLM Systems & Scaling:** Led 400M-parameter Transformer pretraining (40B tokens on 8×H100), delivering reproducible evaluation pipelines to support model selection and deployment decisions for downstream LLM applications (e.g., retrieval, reasoning, and user-facing AI systems) under compute constraints.
- **System Optimization & Deployment Readiness:** Profiled and resolved training/inference bottlenecks (GPU utilization, memory, latency). Optimized trade-offs to improve stability, throughput, and production readiness for downstream LLM applications.

The University of Western Ontario 09/2021 – 07/2025

Graduate Research Assistant *London, ON*

- **Production Semantic Retrieval:** Built scalable retrieval system over 500K+ documents / 1.7M passages, reducing latency by 5× while improving ranking quality, enabling faster and more reliable responses in downstream LLM applications.
- **Evaluation & Experimentation:** Designed and ran large-scale experiments to evaluate retrieval and generation quality, including ablation studies and benchmarking pipelines, identifying failure modes and enabling data-driven improvements in model robustness and performance.

Selected Projects

CineSeek: Production-Style Semantic Movie Search System 2026

Live Demo | GitHub

- **Retrieval-Augmented GenAI System:** Designed a production-style retrieval and ranking system integrating query rewriting, FAISS-based ANN retrieval, and LLM reranking, enabling low-latency, high-quality search for complex natural language queries and improving answer relevance, robustness, and overall user experience.
- **Agent-Based Query Understanding:** Developed an LLM-based agent for query understanding and explanation, improving interpretability and reducing failure cases in user-facing search interactions.
- **System Serving & Deployment:** Built a FastAPI backend with real-time query serving and interactive UI, containerized with Docker and deployed as an HTTPS-enabled service, with a CI/CD pipeline (GitHub Actions) enabling automated build and continuous deployment on each update, ensuring stable access and rapid iteration.

Skills

- **LLM Product & Evaluation**: RAG systems, embedding retrieval, A/B testing, offline/online evaluation, failure analysis (hallucination, retrieval errors), prompt engineering, model quality benchmarking.
- **LLM Systems & Training**: Pretraining, inference optimization (KV cache, batching, vLLM), post-training, transformer architectures, DeepSpeed ZeRO, mixed precision.
- **Production Infrastructure**: FastAPI, Docker, model serving, scalable pipelines, FAISS, LangChain, OpenAI API, SLURM, AWS, Git. Python, PyTorch, HuggingFace Transformers.

Education

The University of Western Ontario 09/2021 – 07/2025

Ph.D. in Computer Science (Natural Language Processing) *London, ON*

- First-author publications in TACL, NAACL, and AAAI (Google Scholar).

The University of Western Ontario 09/2019 – 08/2021

Master of Science in Computer Science (NLP & Machine Learning) *London, ON*